

Review Exercises

Problem 9

An unbiased die is rolled 3 times. Let U , V , and W be the outcomes of the first, second, and third rolls respectively. Define the random variable

$$Z = \max\{U, V, W\}.$$

Find the probability mass function (PMF) of Z .

Problem Statement

Suppose an unbiased six-sided die is rolled three times. Let:

- U : outcome of the first roll
- V : outcome of the second roll
- W : outcome of the third roll

Define the random variable

$$Z = \max\{U, V, W\}.$$

Here we are interested to find the PMF of the discrete random variable Z .

Step-by-Step Solution:

Each die roll is independent and uniformly distributed on $\{1, 2, 3, 4, 5, 6\}$. Therefore, for each of U, V, W , we have:

$$P(U = k) = \frac{1}{6}, \quad k = 1, 2, \dots, 6.$$

Step 1: Understand the distribution of $Z = \max\{U, V, W\}$

We use the cumulative distribution function (CDF) approach:

$$P(Z \leq z) = P(U \leq z, V \leq z, W \leq z).$$

Since the die rolls are independent:

$$P(Z \leq z) = P(U \leq z)^3 = \left(\frac{z}{6}\right)^3.$$

Step 2: Compute the PMF using the CDF

$$P(Z = z) = P(Z \leq z) - P(Z \leq z - 1).$$

Step 3: Compute values for $z = 1, 2, 3, 4, 5, 6$

z	$P(Z \leq z) = \left(\frac{z}{6}\right)^3$	$P(Z \leq z - 1)$	$P(Z = z)$
1	$\left(\frac{1}{6}\right)^3 = \frac{1}{216}$	0	$\frac{1}{216}$
2	$\left(\frac{2}{6}\right)^3 = \frac{8}{216}$	$\frac{1}{216}$	$\frac{7}{216}$
3	$\left(\frac{3}{6}\right)^3 = \frac{27}{216}$	$\frac{8}{216}$	$\frac{19}{216}$
4	$\left(\frac{4}{6}\right)^3 = \frac{64}{216}$	$\frac{27}{216}$	$\frac{37}{216}$
5	$\left(\frac{5}{6}\right)^3 = \frac{125}{216}$	$\frac{64}{216}$	$\frac{61}{216}$
6	1	$\frac{125}{216}$	$\frac{91}{216}$

Step 4: Final Answer

$$P(Z = z) = \begin{cases} \frac{1}{216}, & z = 1 \\ \frac{7}{216}, & z = 2 \\ \frac{19}{216}, & z = 3 \\ \frac{37}{216}, & z = 4 \\ \frac{61}{216}, & z = 5 \\ \frac{91}{216}, & z = 6 \\ 0, & \text{otherwise} \end{cases}$$

Verification: Sum of probabilities should equal 1:

$$\frac{1 + 7 + 19 + 37 + 61 + 91}{216} = \frac{216}{216} = 1 \quad \checkmark$$

Problem 19

Let $X \sim \text{Uniform}(0, 2)$ and $Y \sim \text{Uniform}(0, 3)$ be independent random variables. Define

$$Z = X + Y.$$

Find the probability density function (PDF) of Z .

Step by Step Solution:

Step 1: PDFs of X and Y

Because X and Y are uniform on their intervals,

$$f_X(x) = \begin{cases} \frac{1}{2}, & 0 \leq x \leq 2, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$f_Y(y) = \begin{cases} \frac{1}{3}, & 0 \leq y \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

Step 2: Support of Z

The sum $Z = X + Y$ takes values in the interval:

$$\text{minimum} = 0 + 0 = 0, \quad \text{maximum} = 2 + 3 = 5.$$

So,

$$Z \in [0, 5].$$

Step 3: Formula for PDF of the sum (convolution)

The PDF of the sum of two independent random variables is given by the convolution:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx.$$

Since $f_X(x)$ and $f_Y(y)$ are zero outside their supports, the limits of integration shrink to where both densities are nonzero:

$$x \in [0, 2], \quad y = z - x \in [0, 3].$$

The second condition implies:

$$z - x \in [0, 3] \implies x \in [z - 3, z].$$

Therefore, the integration limits become the intersection:

$$x \in [0, 2] \cap [z - 3, z].$$

We want the interval:

$$[\max(0, z - 3), \min(2, z)].$$

Step 4: Detailed case analysis of the limits

Recall that the integration variable x must satisfy both:

$$0 \leq x \leq 2,$$

and

$$z - 3 \leq x \leq z.$$

To find the intersection:

- The lower limit is:

$$a = \max(0, z - 3).$$

- The upper limit is:

$$b = \min(2, z).$$

The PDF is then

$$\begin{aligned} f_Z(z) &= \int_a^b f_X(x) f_Y(z - x) dx \\ &= \int_a^b \frac{1}{2} \cdot \frac{1}{3} dx = \frac{1}{6}(b - a), \end{aligned}$$

provided $a \leq b$. Otherwise, the integral is zero.

Find a and b for different ranges of z :

1. ****When $z < 0$:****

$$a = \max(0, z - 3) = 0, \quad b = \min(2, z) = z < 0,$$

so $a > b$. Hence,

$$f_Z(z) = 0.$$

2. ****When $0 \leq z < 2$:****

- Compute:

$$a = \max(0, z - 3) = 0 \quad (\text{since } z - 3 < 0 \text{ for } z < 2),$$

$$b = \min(2, z) = z.$$

Since $a = 0 \leq z = b$, the limits are valid.

$$f_Z(z) = \frac{1}{6}(z - 0) = \frac{z}{6}.$$

3. ****When $2 \leq z < 3$:****

- Compute:

$$a = \max(0, z - 3) = 0 \quad (\text{since } z - 3 < 0 \text{ for } z < 3),$$

$$b = \min(2, z) = 2.$$

Limits valid: $a = 0 \leq 2 = b$.

$$f_Z(z) = \frac{1}{6}(2 - 0) = \frac{1}{3}.$$

4. ****When $3 \leq z \leq 5$:****

- Compute:

$$a = \max(0, z - 3) = z - 3 \quad (\geq 0),$$

$$b = \min(2, z) = 2 \quad (\leq z).$$

Limits valid if $a \leq b$:

$$z - 3 \leq 2 \iff z \leq 5,$$

which holds for this range.

$$f_Z(z) = \frac{1}{6}(2 - (z - 3)) = \frac{1}{6}(5 - z).$$

5. ****When $z > 5$ ****

$$a = \max(0, z - 3) = z - 3 > 2,$$

$$b = \min(2, z) = 2.$$

So $a > b$ and the integral is zero.

Step 5: Summary of the PDF $f_Z(z)$

$$f_Z(z) = \begin{cases} 0, & z < 0, \\ \frac{z}{6}, & 0 \leq z < 2, \\ \frac{1}{3}, & 2 \leq z < 3, \\ \frac{5-z}{6}, & 3 \leq z \leq 5, \\ 0, & z > 5. \end{cases}$$

Step 6: Check that total area is 1

$$\int_{-\infty}^{\infty} f_Z(z) dz = \int_0^2 \frac{z}{6} dz + \int_2^3 \frac{1}{3} dz + \int_3^5 \frac{5-z}{6} dz = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1.$$

Total area under $f_Z(z)$ is 1, confirming it is a valid PDF.